Enrolment No./Seat No_______________

GUJARAT TECHNOLOGICAL UNIVERSITY

BE - SEMESTER–IV (NEW) EXAMINATION – SUMMER 2024

Subject Code:3140702 **Date:15-07-2024**

Subject Name: Operating System

Time:10:30 AM TO 01:00 PM **Total Marks:70**

Instructions:

1. **Attempt all questions.**
2. **Make suitable assumptions wherever necessary.**
3. **Figures to the right indicate full marks.**
4. **Simple and non-programmable scientific calculators are allowed.**

MARKS

Q.1 **(a)** Define concurrency in OS and briefly explain its associated challenges. **03**

(b) Explain "5 State" process state transition diagram with an illustration. **04**

(c) Discuss the variety of task types that the operating system performs. **07**

Q.2 **(a)** Define the following terms: **03**

i) Critical Section
ii) Race Condition
iii) Mutual Exclusion

(b) Differentiate Multiprocessing and Multithreading. **04**

(c) **07**

Process	Arrival Time	Processing Time
A	0	3
B	1	6
C	4	4
D	6	2

Calculate the average waiting and turnaround time for the processes listed in the table using the following scheduling algorithms: the CPU burst time length in milliseconds.

i) First Come First Serve
ii) Non-preemptive Shortest Job First
iii) Shortest Remaining Time Next
iv) Round Robin with Quantum value two

OR

(c) **07**

Process	Burst Time	Priority
P1	10	3
P2	1	1
P3	2	3
P4	1	4
P5	5	2

Calculate the average waiting and turnaround time for the processes listed in the table using the following scheduling algorithms: the CPU burst time length in milliseconds. Consider 1 as the highest priority.

i) Preemptive Shortest Job First
ii) Non - preemptive Priority Scheduling
iii) Preemptive - Priority Scheduling
iv) Round Robin with Quantum value one

Q.3 **(a)** Define and explain Message Passing. **03**

(b) Provide a brief description of the following: **04**

i) Starvation
ii) Aging

	(c)	What is a scheduler? Illustrate queuing with an appropriate diagram.	**07**

OR

Q.3	**(a)**	Differentiate the following: i) Semaphore ii) Monitor	**03**
	(b)	Provide a brief description of the following: i) Deadlock Prevention ii) Deadlock Avoidance	**04**
	(c)	Describe Peterson's Algorithm for Process Synchronization.	**07**
Q.4	**(a)**	Compare Multiprogramming with Fixed Partition and Multiprogramming with Variable Partition.	**03**
	(b)	Define Authentication. How can operating system security be ensured using authentication?	**04**
	(c)	1. What is Stripping in RAID? Give an appropriate example. 2. Suppose that a disk drive has 200 cylinders. The drive currently serves at cylinder 50. The queue of pending requests in FIFO order is 82, 170, 43, 140, 24, 16, 190. Starting from the current head position, what is the total distance (in cylinders) the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms? i) FCFS ii) SCAN iii) C-SCAN	**07**

OR

Q.4	**(a)**	What is TLB(Translation Lookaside Buffer), and what is the purpose of it?	**03**
	(b)	Discuss the Access Control List in brief.	**04**
	(c)	1. What is Mirroring in RAID? Give an appropriate example. 2. Suppose that a disk drive has 200 cylinders. The drive currently serves at cylinder 100; the previous request was at cylinder 81. The queue of pending requests in FIFO order is 55,58,39,18,90,160,150,38,184. Starting from the current head position, what is the total distance (in cylinders) the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms? i) SSTF ii) LOOK iii) C-LOOK	**07**
Q.5	**(a)**	Enlist the advantages and disadvantages of Virtual Machines.	**03**
	(b)	Write a shell script to display all executable files, directories, and zero-sized files from the current directory.	**04**
	(c)	Consider the reference string: 4, 7, 6, 1, 7, 6, 1, 2, 7, 2. The number of frames in the memory is 3. Find out the number of page faults respective to: i) Optimal Page Replacement Algorithm ii) FIFO Page Replacement Algorithm iii) LRU Page Replacement Algorithm	**07**

OR

Q.5	**(a)**	Describe virtualization with an appropriate diagram.	**03**
	(b)	Advantages of LINUX/UNIX operating system over Windows.	**04**

(c) Given memory partition of 100K, 500K, 200K, 300K, and 600K in order. How would each of the First-fit, Best-fit and Worst-fit algorithms place the processes of 212K, 417K, 112K and 426K in order? Which algorithm makes the most efficient use of memory? Show the diagram of memory status in each case. **07**

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